Grade 9 Mathematics – Loci and Constructions

Tutorial – Structured (Essay-type) Questions

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Name: ..... Date: .....

Instructions: Answer all questions. Use a sharp pencil, a straight edge and a pair of compasses. Do not rub out your construction arcs. Use a protractor only where the question says so.

Question 1

By a rough sketch, show the locus in each situation below and name the type of locus.

(i) A goat is tied to a peg P with a rope of length 3 m. The path of the goat when it moves around keeping the rope stretched.

(ii) The path of the tip of the minute hand of a clock, which is 10 cm long.

(iii) The path of a boy who walks so that he is always at an equal distance from two trees A and B which are 20 m apart.

(iv) The path of the centre of a wheel of diameter 40 cm rolling along a straight, flat road.

(v) The path of a ball rolling on the floor so that it is always at an equal distance from two straight walls OA and OB which meet at the corner O.

Question 2

Use only a straight edge and a pair of compasses.

(i) Draw the line segment AB such that AB = 8 cm.

(ii) Construct the perpendicular bisector of AB.

(iii) Construct the locus of points which are 5 cm from A, and name the points where it meets the perpendicular bisector as C_1 and C_2.

(iv) Measure and write the length of BC_1 and the distance from C_1 to AB.

Question 3

P and Q are two trees on flat ground, 10 m apart. A tap is to be fixed 6 m from P and 8 m from Q. Use the scale 1 cm = 1 m.

(i) Draw PQ to scale.

(ii) Using your knowledge of loci, find and mark the possible positions of the tap as T_1 and T_2.

(iii) Measure and write the magnitude of the angle PT_1Q.

Question 4

PQ is a straight drain of length 50 m. A tap is to be fixed at a distance of 10 m from PQ and at an equal distance from both ends P and Q. Use the scale 1 cm = 5 m.

(i) Draw PQ to scale.

(ii) Construct the locus of points which are 10 m from PQ.

(iii) Construct the locus of points which are equidistant from P and Q.

(iv) Mark the possible positions of the tap. How many such positions are there?

(v) Measure the distance from P to one of these positions and find the actual distance.

Question 5

PQRS is a rectangular plot of land with PQ = 30 m and PS = 20 m. A tree is to be planted 8 m from the boundary PQ and 5 m from the boundary PS.

(i) Draw a rough sketch of the plot and show the two relevant loci.

(ii) Mark the position where the tree should be planted and name it T.

(iii) Write the distances from T to the boundaries SR and QR.

Question 6

(i) Draw a straight line AB such that AB = 7 cm, and mark the point P on AB such that AP = 3 cm.

(ii) Construct a line perpendicular to AB through P.

(iii) Mark the point Q on the perpendicular line such that PQ = 4 cm.

(iv) Measure and write the length of AQ.

Question 7

(i) Using a protractor, draw the angle $X\hat{Y}Z = 120°$, mark YX = YZ = 5 cm on its arms and complete the triangle XYZ.

(ii) Construct a perpendicular line from Y to XZ and name the point at which it meets XZ as D.

(iii) Measure and write the lengths of XD, ZD and YD.

Question 8

Use only a straight edge and a pair of compasses.

(i) Draw a straight line segment PQ of length 8 cm.

(ii) Construct angles of 60° at P and at Q on the same side of PQ. Name the point where the arms meet as R.

(iii) Measure PR and QR, and name the type of triangle PQR.

(iv) Construct the bisector of the angle $Q\hat{P}R$ and name the point where it meets QR as S.

(v) Measure and write the length of QS and the magnitude of $P\hat{S}Q$.

Question 9

Construct the following angles using only a straight edge and a pair of compasses.

(i) Draw AB = 6.5 cm and construct an angle of 90° at A such that AB is an arm. Then bisect it to obtain an angle of 45°.

(ii) Draw OA = 6 cm and construct an angle of 60° at O. Bisect it to obtain 30°, and bisect again to obtain 15°.

(iii) Construct an angle of 75°. (Hint: 75° lies exactly halfway between 60° and 90°.)

Question 10

To construct a triangle PQR, carry out the following steps.

(i) Draw a straight line segment PQ of length 7 cm.

(ii) Construct an angle of 30° at P such that PQ is an arm.

(iii) Construct an angle of 45° at Q such that QP is an arm.

(iv) Complete the triangle PQR. Measure and write the magnitude of $P\hat{R}Q$ and the lengths of PR and QR.

Question 11

(i) Draw any acute angle with LM = 6 cm as one arm and name it K$\hat{L}$M.

(ii) Copy the angle $\hat{L}$ at M such that K$\hat{L}$M = L$\hat{M}$N, where N is on the same side of LM as K.

(iii) Name the point of intersection of the lines LK and MN as P (produce the lines if necessary). Measure and write the lengths of PL and PM.

(iv) What type of triangle is PLM?

Question 12

(i) Draw a circle of radius 4 cm and name its centre O.

(ii) Mark two points A and B on the circle 6 cm apart from each other, and draw the chord AB.

(iii) Construct a perpendicular line from O to AB and name the point at which it meets AB as N.

(iv) Measure and write the lengths of AN, BN and ON. What do you observe?

Question 13

A rope is stretched tightly between two posts A and B which are 10 m apart. A ring that can slide freely along this rope is tied to a goat with another rope of length 4 m.

(i) Draw a rough sketch, with measurements, of the boundary of the region the goat can reach.

(ii) Describe the parts that make up this boundary (locus).

(iii) Find the area of the region on which the goat can graze. (Use $\pi = 22/7$.)